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Farm water supply: plastic-lined catchments

Plastic-lined catchments are best suited to the flat, low-rainfall areas of the WA wheatbelt, where they can improve farm water security by increasing runoff capture. They should be considered alongside options such as increasing storage, improving existing catchments or infrastructure to improve water use efficiency.

Plastic liners can be used on both catchments and dams. This fact sheet focuses on catchment lining, although dam lining may also be worthwhile where seepage losses or water quality issues occur.

A dam is only as good as the catchment that fills it. By reducing infiltration losses, plastic-lined catchments can generate runoff from rainfall events as small as 1 mm, compared with many roaded catchments that typically require 8–10 mm of rain before runoff.

This fact sheet covers catchment materials, installation, sizing and some WA examples.

Considering a plastic-lined catchment

Step 1: Know your water requirements

Before investing in a plastic-lined catchment consider:

- How much water is required for livestock, spraying and other farm operations?
- Are you in a low-rainfall area where events greater than 10 mm are uncommon?
- Could improving an existing roaded catchment provide the water you need?

Step 2: Start with your dam

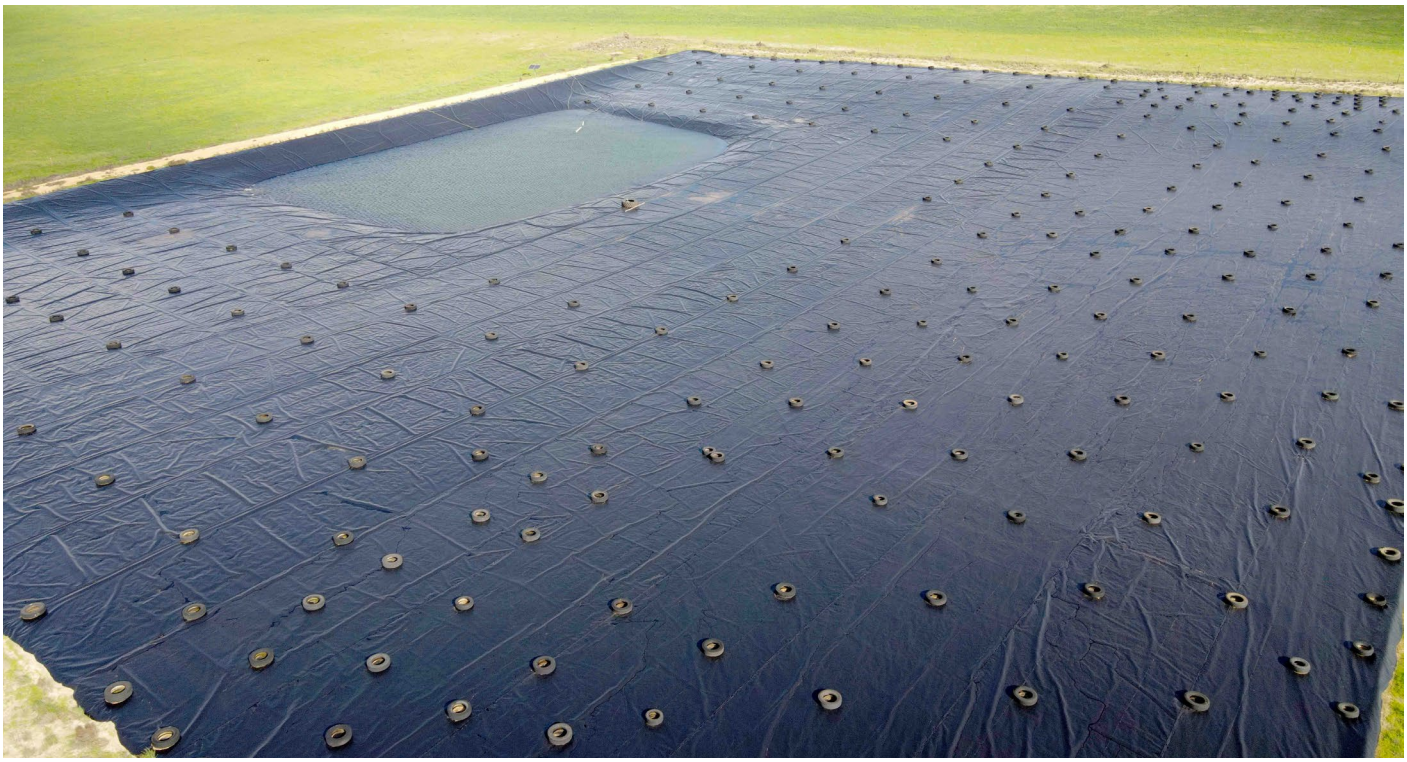
A plastic-lined catchment can significantly increase runoff, so make sure your dam is ready to receive it.

- Is the dam large enough to store the additional water?
- Does it need cleaning, desilting or enlargement?
- Is the spillway designed to safely convey peak flows and overflow?

  Correct sizing is critical: an undersized spillway is a common cause of dam failure, because water that can't bypass the dam overtops the embankment, eroding and breaching the wall. Size the spillway to safely carry the peak discharge.

Step 3: Size the catchment to match your needs

Match the size of your modified runoff catchment (for example, plastic-covered areas, roads or other hard surfaces) to your water demand, dam capacity and local climate. Bigger is not always better. Information on sizing a catchment is provided at the end of this fact sheet.



A lined dam and catchment in South Australia. The liner is weighted with tyres. [Source: South Australian Research and Development Institute]

Things to know about plastic liners

Material: Plastic liners come in a range of qualities including High Density Polyethylene (HDPE), Linear Low-Density Polyethylene (LLDPE), Polypropylene (PP) and Poly Vinyl Chloride (PVC), each with different properties. HDPE geomembranes are the most common liner used for waterproofing lined channels and dams, because their high strength to density ratio gives excellent durability and UV and chemical resistance.

Thickness: 1.5 mm is a common choice for lined catchments. For runoff you want a durable, smooth surface that lasts. HDPE geomembranes come in a range of thicknesses, for catchments and dams, 1–2 mm is usual. Thicker liners give more puncture resistance and durability but cost more and can raise handling and installation considerations which your supplier can advise on.

UV resistance and liner longevity: Because catchment liners are exposed to sunlight throughout their working life, UV resistance is critical. HDPE liners rely on carbon black, incorporated throughout the polyethylene during manufacture, to absorb and block UV radiation, helping maximise durability, performance and service life. Most plastic liners are black because carbon black provides the primary UV protection. White reflective liners use the same UV-protected black HDPE core with a white outer layer. They may run cooler in hot conditions, but long-term Australian performance data is limited.

Industry specification for HDPE geomembranes: GRI-GM13 is the recognised industry standard for HDPE geomembranes. To meet this standard, liners must pass a range of quality and durability tests, including UV resistance and long-term ageing performance.

Product formulation and performance may vary between manufacturers, so growers should ask suppliers to provide evidence of GRI-GM13 compliance.

Site preparation and installation

Earthworks: A gentle slope (typically 2–3%) is sufficient to direct runoff to the dam. The catchment should be cleared of sharp objects, with any necessary compaction, and optional underlay (sand, geomembranes etc.) completed to protect the liner. Existing roaded catchments will likely require regrading to achieve suitable liner slopes and a smoother surface.

Fencing: Lined catchments should be fenced to exclude livestock and wildlife and minimise liner damage.

Exit points: Where dams are also lined, fencing and safe exit points are important safety measures.

Installation:

- HDPE liners are supplied in rolls and joined on-site using plastic welding.
- Seam quality is critical. Ask installers how welds will be tested and quality checked.
- Liners are commonly secured with an anchor trench around the perimeter and weighted across the catchment with tyres, tyre sidewalls or plastic sandbags.
- Large sheets can catch the wind, so installation requires care and appropriate safety measures.
- Most suppliers can provide or recommend experienced installation contractors.
- Good site preparation, welding and anchoring are essential for long-term liner performance.



The quality of on-site seam welding is critical

Protect the dam inlet from erosion: Fast-moving runoff from plastic-lined catchments can scour soil and undermine the liner. Where possible, extend the liner to the dam base (or below the minimum water level), ensuring the inlet is protected. Alternatively, install erosion protection such as rock armouring, or concrete composite mats. Concrete Composite Mats are roll-out products that harden when wet to form a durable concrete surface. They can be used to line drains, channels and dam inlets but are generally heavier and more expensive than plastic liners.



Anchor trench at an installation at Newdegate before infilling

Indicative costs

Plastic liner:	Approximately $60,000–$80,000 (inc. GST) for a 1 ha catchment of 1.5 mm HDPE. Excludes freight, earthworks, fencing and installation.
Freight, lead times and minimum order quantities:	Check these early, as they can significantly affect project costs and delivery schedules. Actual liner lead times can be significantly longer than the quoted 8–10 weeks.
Earthworks and fencing:	Costs vary considerably depending on site conditions and local contractor rates.
Installation:	Contractor installation costs vary, but budget approximately $3/m² (around $30,000/ha) for liner installation. Confirm mobilisation and standby charges upfront, as these can significantly affect total project costs.
Very approximate total project cost:	Around $120,000–130,000 for 1 ha, including liner supply, freight, earthworks, fencing and installation. Actual costs will vary between sites, and liner prices may increase over time due to changes in material and freight costs. Plastic costs are linked to global oil prices.

Planning and approvals for farm water storages

Before building or enlarging a dam confirm any approval requirements with your local shire.

South Australia

Plastic liners for both catchment and dam have seen widespread adoption in South Australia. You can read more information in the Water Harvesting and Lined Catchment fact sheet.



A South Australian example of lined catchment and dam [Source: South Australian Research and Development Institute]

Examples of WA installations

Newdegate: white plastic catchment

Catchment area:	Approximately 1.2–1.6 ha
Liner assessed:	1.5 mm, 2.0 mm and 2.5 mm HDPE (black and white reflective options)
Liner selected:	1.5 mm white/black HDPE
Additional protection:	2.0 mm black HDPE liner installed at the dam inlet to reduce erosion risk
Earthworks:	Approximately 15–16 hours of grading to prepare the existing roaded catchment, plus ~5 hours of loader work for the inlet channel
Installation:	Approximately 210 labour hours to roll out, weld and secure the liner



Example of a white plastic catchment near Newdegate

Newdegate: two black plastic catchments

Catchment area:	Installed two catchment areas 2–2.4 ha
Liner selected:	Catchment 1 was 1.5 mm black HDPE and catchment 2 was 1 mm HDPE
Additional protection:	Liner installed into the sediment trap on the dams
Installation:	Contractor installed



Example of a black plastic catchment near Newdegate

Dunn Rock: lined dam and catchment

Lining both the catchment and dam may improve spray water quality and reliability where dispersive clay soils cause persistently milky dam water. Where dams are lined, fencing and safe exit points are important safety requirements.



Example of a WA lined dam and catchment. The catchment is ~0.7–1 ha

Borden: black plastic catchment

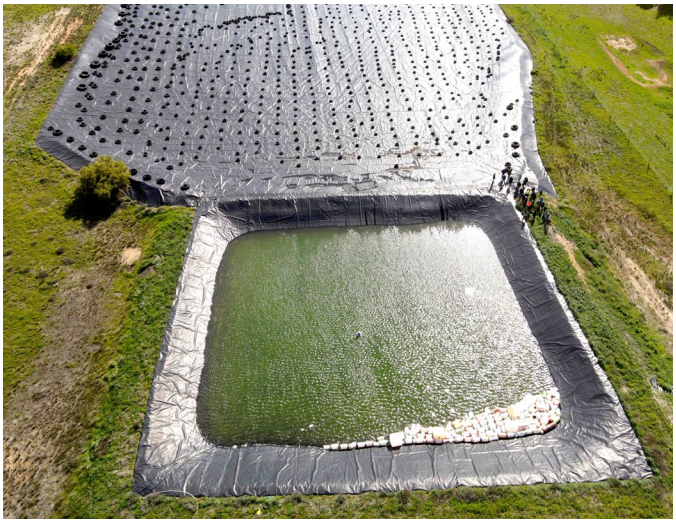
Catchment area:	5 ha
Liner selected:	1 mm Black HDPE
Additional protection:	Liner extended into dam at delivery point
Installation:	Contractor installed



Large HDPE lined catchment at Borden during construction

WA grower learnings from farm installations

- **Design the whole system.** Consider dam capacity, tanks, spillways and inlet erosion protection before installing a lined catchment.
- **Size the catchment to your needs.** Match catchment area to dam capacity, water demand and local climate.
- **Bigger isn't always better.** Plastic-lined catchments are highly efficient at generating runoff, so a smaller catchment may achieve your water supply goals at lower cost.
- **Understand all contractor costs upfront.** Confirm installation rates, mobilisation, demobilisation, accommodation and any standby or weather-delay charges before work begins.
- **Allow extra time for liner delivery.** Lead times can be significantly longer than quoted, particularly during periods of high demand, so order early and plan accordingly.
- **Consider tanks, pumps or additional storage at dam site prior to installation.**



A South Australian lined catchment and dam [Source: South Australian Research and Development Institute]

How big should my catchment be?

Water Evaluation Platform – recommended approach

As part of the WaterSmart project, the Water Evaluation Platform (WEP) helps growers estimate farm water demand and assess dam and catchment requirements using local climate data.

By combining your water use with historical rainfall and evaporation records, WEP can simulate dam performance over time, helping you determine the most suitable dam and catchment size for your property.



Back-of-envelope calculations

Use the following calculations as a starting point only. For detailed assessment, use WEP.

1. Estimate your water demand

Livestock: Use DPIRD livestock water requirement guides.

Spraying: Annual spray water demand (L) = 80–120 L/ha × cropped area (ha) × number of spray applications + extra water for tank, boom and line cleaning.

2. Estimate catchment runoff

A simple estimate of annual runoff is:

Runoff (kL) = Catchment Area (m²) × Rainfall (mm) × Runoff Fraction ÷ 1000

For plastic-lined catchments, use a runoff fraction of:

- 0.8 for conservative water security planning
- 0.9–0.95 when sizing storage capacity.

Example:

A 1.5 ha catchment (15,000 m²) receiving 300 mm rainfall:

(15,000 × 300 × 0.9) ÷ 1000 = 4,050 kL (4.05 ML)

Conversions: 1,000 L = 1 kL 1,000 kL = 1 ML

3. Don’t forget evaporation loss

Annual evaporation across the WA Wheatbelt ranges from around 1 to more than 2.5 metres per year, based on Bureau of Meteorology average annual pan evaporation data. Rather than relying on regional averages, the Water Evaluation Platform (WEP) uses location-specific gridded climate data to estimate evaporation losses and show how much water is lost from your dam over time.



4. Peak discharge and spillway design: seek expert advice

An undersized spillway is one of the most common causes of dam failure. Plastic-lined catchments can generate large volumes of runoff very quickly, placing additional pressure on dam inlets, spillways and embankments.

Don’t rely on a back-of-the-envelope calculation. Peak flows depend on local rainfall intensity, catchment size and dam design. Seek advice from an experienced engineer or earthworks contractor when designing or modifying a dam. The cost of getting the design right is small compared with the cost of repairing a failed dam.

WaterSmart Farms

For more on farm water supply and security options, see WaterSmart Farms, DPIRD’s suite of projects building water security in a drying climate. Have questions about any of the WaterSmart Farms projects? Contact the DPIRD Water Science team at water.science@dpird.wa.gov.au.

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